



LINES & ANGLES

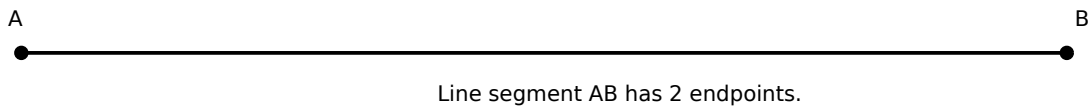
Complete Class 6 → Class 7 → Class 9 Notes with Visual Diagrams

Every major concept has a simple figure so you can connect the definition with the picture.

1. Basic Geometry: Point, Line, Segment and Ray

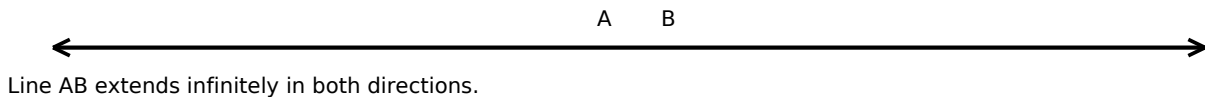
Point

A point represents an exact position. It has no length or width. It is usually named by a capital letter such as A.



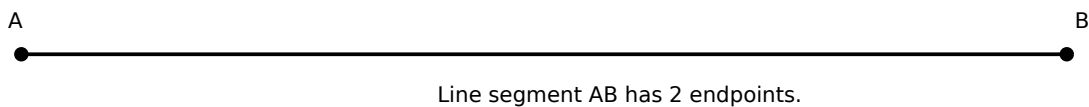
Line

A line is straight and extends endlessly in both directions.



Line Segment

A line segment is a part of a line with exactly two endpoints.



Ray

A ray has exactly one endpoint and extends endlessly in one direction.

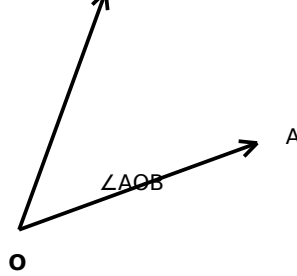


Memory trick

Line = 0 endpoints • Ray = 1 endpoint • Segment = 2 endpoints

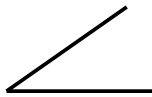
2. Angles — Meaning and Parts

An angle is formed by two rays with a common endpoint. The common endpoint is the vertex; the rays are the arms or sides.

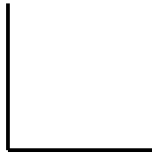


- $\angle AOB$ is read as angle AOB. The middle letter O is the vertex.
- An angle can be measured in degrees ($^\circ$).
- The amount of turn from one arm to the other determines the angle measure.

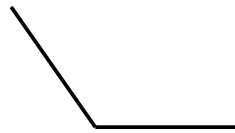
Types of angles



Acute



Right



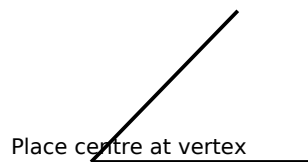
Obtuse



Straight

Type	Range / value	Key idea
Zero	0°	No turn
Acute	$0^\circ < \theta < 90^\circ$	Less than 90°
Right	90°	Quarter turn
Obtuse	$90^\circ < \theta < 180^\circ$	Between 90° and 180°
Straight	180°	Half turn / straight line
Reflex	$180^\circ < \theta < 360^\circ$	More than a straight angle
Complete	360°	One full turn

3. Measuring Angles with a Protractor



- Put the centre of the protractor exactly on the vertex.
- Put the baseline along one arm.
- Use the scale that starts at 0° on that arm.
- Read where the second arm crosses the scale.
- Always write the degree symbol.

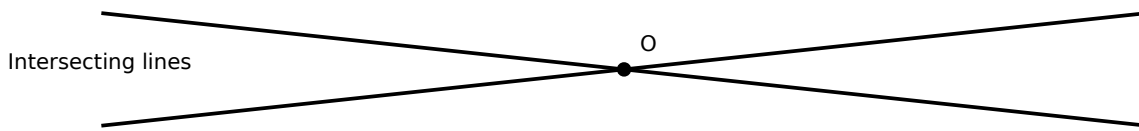
Common mistake

A protractor has two scales. Do not simply choose the first number you see; choose the scale whose 0° is on the starting arm.

4. Lines: Intersecting, Parallel and Perpendicular

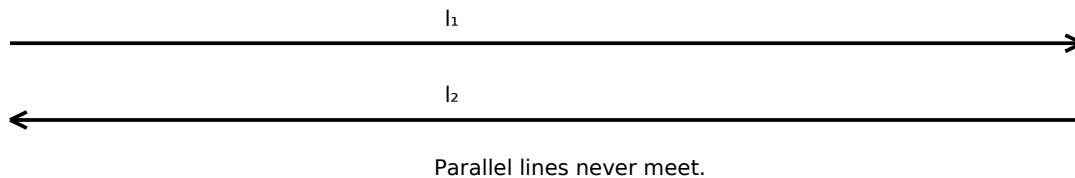
Intersecting lines

Two lines are intersecting if they meet at one point.



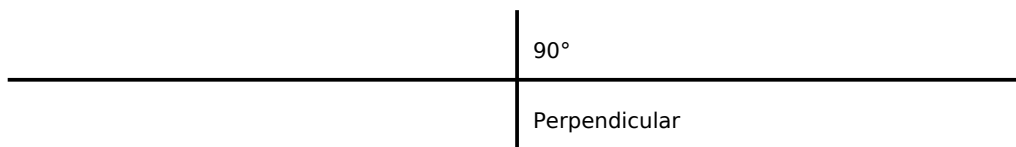
Parallel lines

Two lines in the same plane that never meet, even when extended, are parallel.



Perpendicular lines

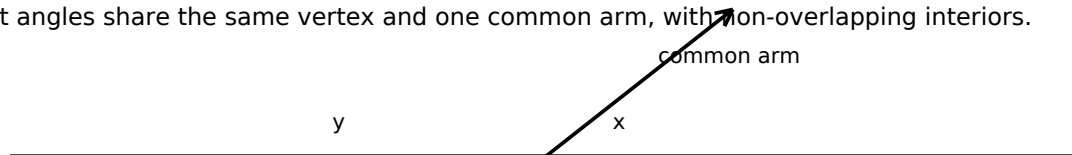
Two lines are perpendicular when they intersect at a right angle.



5. Class 7 — Pairs and Relationships of Angles

Adjacent angles

Adjacent angles share the same vertex and one common arm, with non-overlapping interiors.



Adjacent angles form a straight line $\rightarrow 180^\circ$

Complementary angles

Two angles whose sum is 90° are complementary.

$$\angle A + \angle B = 90^\circ$$

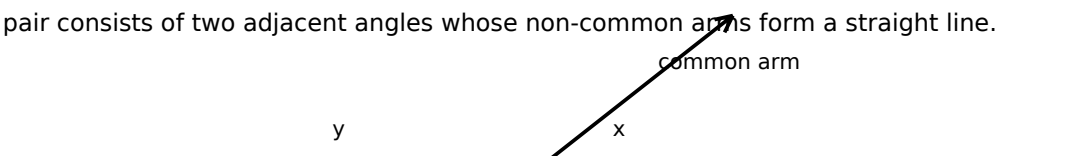
Supplementary angles

Two angles whose sum is 180° are supplementary.

$$\angle A + \angle B = 180^\circ$$

Linear pair

A linear pair consists of two adjacent angles whose non-common arms form a straight line.

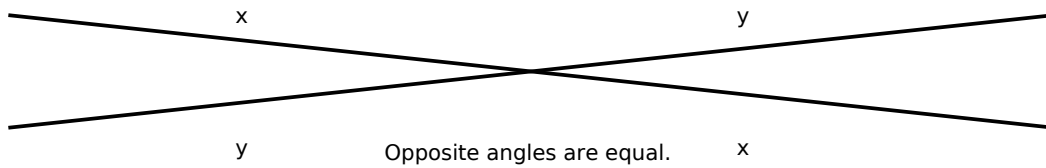


Adjacent angles form a straight line $\rightarrow 180^\circ$

$$\text{Linear pair sum} = 180^\circ$$

Vertically opposite angles

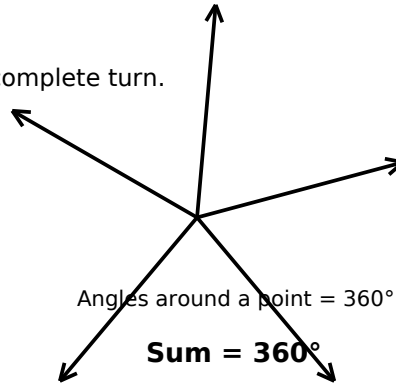
When two lines intersect, opposite angles are vertically opposite. They are always equal.



If one is 70° , the opposite one is 70° ; each adjacent angle is 110° .

Angles around a point

All angles around a point make one complete turn.

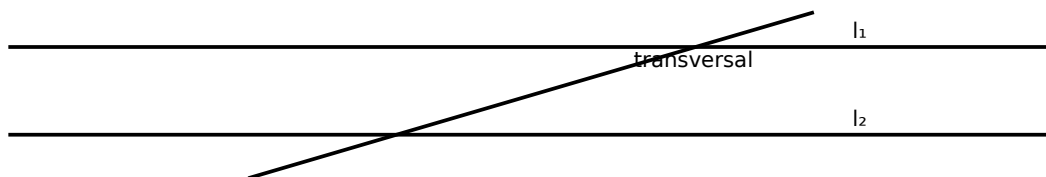


Angles on a straight line

Angles forming a straight line have total measure 180° .

6. Transversal – The Bridge to Class 9

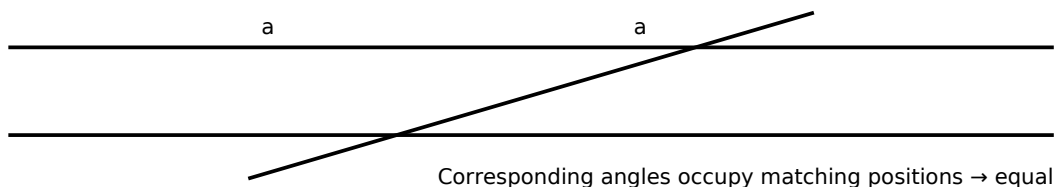
A transversal is a line that intersects two or more lines at distinct points.



When a transversal cuts two parallel lines, it creates special angle pairs.

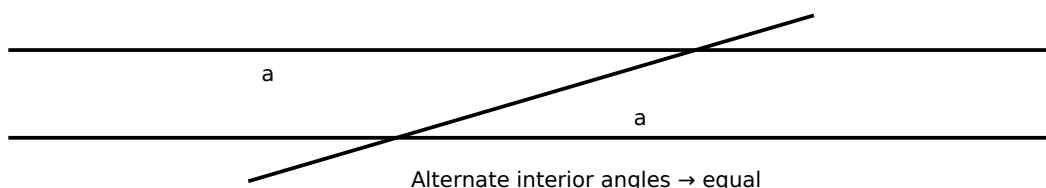
Corresponding angles

They are in matching positions at the two intersections. For parallel lines, corresponding angles are equal.



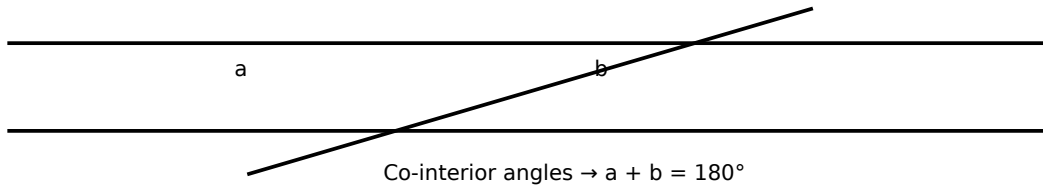
Alternate interior angles

They lie between the parallel lines and on opposite sides of the transversal. They are equal.



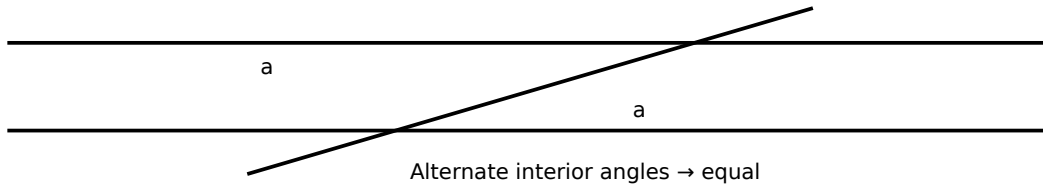
Co-interior / same-side interior angles

They lie between the two lines on the same side of the transversal. Their sum is 180° .



Alternate exterior angles

They lie outside the two lines and on opposite sides of the transversal. For parallel lines, they are equal.



7. Class 9 — Theorem Results

Theorem 1: If a transversal intersects two parallel lines, corresponding angles are equal.

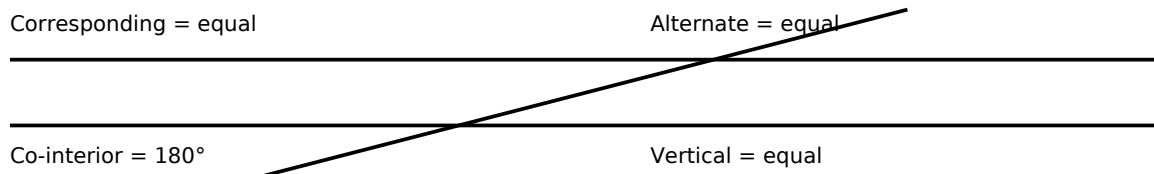
Theorem 2: If a transversal intersects two parallel lines, alternate interior angles are equal.

Theorem 3: If a transversal intersects two parallel lines, interior angles on the same side are supplementary.

Converse: If corresponding angles are equal, the two lines are parallel.

Converse: If alternate interior angles are equal, the two lines are parallel.

Converse: If same-side interior angles have sum 180° , the two lines are parallel.



8. Solving Any Angle Problem

- 1. Mark the known angle.
- 2. Identify the geometric situation.
- 3. Select the rule: 90° , 180° , 360° , vertical, corresponding, alternate, or co-interior.
- 4. Write an equation.
- 5. Solve for the unknown.
- 6. Check the result against the figure.

Example 1 — Linear pair

One angle = 65° , other = x .

$$x + 65^\circ = 180^\circ \rightarrow x = 115^\circ$$

Example 2 — Vertically opposite

One angle = 48° , opposite angle = x .

$$x = 48^\circ$$

Example 3 — Around a point

Angles are 80° , 95° , 70° and x .

$$80 + 95 + 70 + x = 360 \rightarrow x = 115^\circ$$

Example 4 — Co-interior

One co-interior angle = 112° .

$$112 + x = 180 \rightarrow x = 68^\circ$$

Example 5 — Algebra

Angles are $(2x + 10)^\circ$ and $(3x - 20)^\circ$ and form a linear pair.

$$(2x + 10) + (3x - 20) = 180 \rightarrow 5x = 190 \rightarrow x = 38.$$

9. Important Differences

Concept	How to identify it
Adjacent	Common vertex + common arm + no overlap
Linear pair	Adjacent + non-common arms form a straight line
Vertically opposite	Opposite angles made by two intersecting lines
Corresponding	Same relative position at two intersections
Alternate interior	Inside the lines + opposite sides of transversal
Co-interior	Inside the lines + same side of transversal
Parallel	Never meet
Perpendicular	Meet at 90°

10. Complete Revision Map

$90^\circ \rightarrow$ complementary & right angle

$180^\circ \rightarrow$ straight line, supplementary, linear pair

$360^\circ \rightarrow$ full turn / angles around a point

Vertical $\rightarrow$ equal

Parallel + corresponding $\rightarrow$ equal

Parallel + alternate $\rightarrow$ equal

Parallel + co-interior $\rightarrow 180^\circ$

11. Practice Questions

- Classify 35° , 90° , 125° , 180° and 250° .
- Find x if x and 72° form a linear pair.
- Vertically opposite angles are x and $(3x - 40)^\circ$. Find x .
- Around a point: 75° , 105° , 80° and x . Find x .
- Complementary angles: $(x + 10)^\circ$ and $(2x + 5)^\circ$. Find x .
- Supplementary angles: $(4x - 10)^\circ$ and $(2x + 40)^\circ$. Find x .
- Corresponding angle = 118° on parallel lines. Find its pair.
- Co-interior angle = 73° . Find the other angle.
- Explain corresponding vs alternate interior angles.
- State three converse conditions that can prove two lines are parallel.

Answers

- Acute, right, obtuse, straight, reflex.
- 108° .
- $x = 20^\circ$.
- 100° .
- $x = 25^\circ$.
- $x = 25^\circ$.
- 118° .

8. 107° .

9. Corresponding = matching positions; alternate interior = inside and opposite sides of transversal.

10. Equal corresponding; equal alternate interior; or co-interior sum 180° .